

Employee Onboarding & HR Management System

Document ID	SRS-33
Domain	Human Resources
Version	1.0
Standard	IEEE Std 830-1998
Stack	Spring Boot 3.x · React.js · JWT · JPA
Status	Draft

Table of Contents

1. Introduction	
1.1 Purpose	
1.2 Scope	
1.3 Definitions, Acronyms, and Abbreviations	
1.4 References	
1.5 Overview	
2. Overall Description	
2.1 Product Perspective	
2.2 Product Functions	
2.3 User Classes and Characteristics	
2.4 Operating Environment	
2.5 Design and Implementation Constraints	
2.6 User Documentation	
2.7 Assumptions and Dependencies	
3. External Interface Requirements	
3.1 User Interfaces	
3.2 Hardware Interfaces	
3.3 Software Interfaces	
3.4 Communications Interfaces	
4. System Features (FR1–FR12)	
4.1–4.12 Functional Requirements	
5. Other Non-Functional Requirements	
5.1 Performance	
5.2 Safety	
5.3 Security	
5.4 Software Quality	
5.5 Database Requirements	
5.6 System Quality Attributes	
5.7 Design Constraints	
5.8 Other Requirements	
Appendix A: Permission Matrix	
Appendix B: Database Schema Details	
Appendix C: Validation Rules	
Appendix D: Security Configuration & Exception Handling	

Appendix E: Integration Specifications	
Appendix F: Business Rules	
Appendix G: Future Roadmap	
Appendix H: API Summary	
Appendix I: React Component Structure	
Version Control History	

1. Introduction

1.1 Purpose

This SRS defines requirements for the Employee Onboarding & HR Management System, a comprehensive human resources platform covering the full employee lifecycle — from pre-joining onboarding through payroll, leave, performance appraisal, and exit. Built on Spring Boot 3.x, React.js, JWT, and JPA, it eliminates paper-based HR processes, ensures statutory compliance (PF, ESI, TDS, PT), and gives HR teams and employees a unified self-service portal.

1.2 Scope

- Pre-joining onboarding: digital offer letter, document collection, and background verification tracking
- Employee master: complete employee profile with org hierarchy, role, and contract details
- Attendance and leave management: biometric/app-based attendance with leave balance tracking
- Payroll processing: automated monthly payroll with statutory deductions and payslip generation
- Performance management: goal setting, mid-year review, and annual appraisal with ratings
- Training and development: training calendar, enrolment, completion tracking, and certification
- Employee self-service: payslip download, leave application, tax declaration, and profile update
- Statutory compliance: PF, ESI, TDS, Professional Tax computation and challan generation
- Exit management: resignation, notice period, clearance, and full-and-final settlement
- HR analytics: attrition, headcount, leave utilisation, and payroll cost dashboards

1.3 Definitions, Acronyms, and Abbreviations

Term	Definition
SRS	Software Requirements Specification
JWT	JSON Web Token
JPA	Java Persistence API
API	Application Programming Interface
PF	Provident Fund
ESI	Employees' State Insurance
TDS	Tax Deducted at Source
PT	Professional Tax
CTC	Cost to Company
LOP	Loss of Pay
KRA	Key Result Area
KPI	Key Performance Indicator
BGV	Background Verification
FnF	Full and Final Settlement
OTP	One Time Password
RBAC	Role-Based Access Control

HRA	House Rent Allowance
YTD	Year-to-Date
HRMS	Human Resource Management System
SLA	Service Level Agreement

1.4 References

- IEEE Std 830-1998
- Companies Act 2013
- Employees Provident Funds Act 1952
- Employees State Insurance Act 1948
- Income Tax Act 1961
- Payment of Wages Act 1936
- Spring Boot 3.x Documentation
- JWT RFC 7519

1.5 Overview

Section 2 describes overall product perspective. Section 3 covers external interfaces. Section 4 details all functional requirements. Section 5 addresses non-functional requirements. Appendices A–I provide supplementary specifications.

2. Overall Description

2.1 Product Perspective

Spring Boot 3.x backend (REST port 8080) + React.js employee self-service (port 8081) + React.js HR console (port 8082). JWT with RBAC: super-admin, HR-manager, department-manager, employee. Razorpay for payroll disbursement. Digio for e-sign on offer letters. PostgreSQL via JPA.

2.2 Product Functions

- Digital pre-joining onboarding with document collection and e-signature
- Employee profile and organisational hierarchy management
- Biometric and mobile app attendance with shift scheduling
- Leave policy configuration and employee leave management
- Automated monthly payroll with CTC breakup and statutory deductions
- Tax declaration (Form 12BB) and TDS computation
- PF, ESI, PT challan generation for statutory filing
- Goal setting, mid-year, and annual performance appraisal
- Training calendar, nomination, completion, and certificate issuance
- Exit workflow: resignation, clearance, and FnF computation
- HR analytics: attrition, headcount, payroll cost, and leave trends
- Employee self-service: payslip, leave application, profile, and IT declaration

2.3 User Classes and Characteristics

Super Admin / HR Manager

Manages all employees, configures HR policies, runs payroll, views all analytics, and handles compliance.

Department Manager

Views team attendance, approves leave and expenses, initiates appraisals, and signs off on exit clearance.

Employee

Self-service: view payslip, apply leave, update profile, declare taxes, complete training, and submit resignation.

2.4 Operating Environment

Backend: Spring Boot 3.x, JDK 17+, PostgreSQL 14+. Frontend: React.js 18.x. Integrations: Biometric device (ZKTeco via SDK), Digio (e-sign), Razorpay (payroll payout), Twilio SMS. Hosted on AWS EC2.

2.5 Design and Implementation Constraints

- PF/ESI/TDS computations must comply with current statutory rates and be auditable
- Payslips must be password-protected PDF (password: DOB in DDMMYYYY format)
- Employee data encrypted at rest (AES-256); payroll data access restricted to HR and finance
- Biometric data (fingerprint templates) not stored in application database; stored in device only
- Spring Boot 3.x, Java 17+; React.js functional components; mobile-responsive self-service portal
- HTTPS-only; JWT auth; RBAC; payroll data access logged in audit trail
- Payroll processing must complete within 2 hours for up to 5,000 employees
- FnF settlement computation must account for all outstanding leaves, deductions, and recoveries

2.6 User Documentation

Employee Self-Service Guide, HR Manager Operations Manual, Payroll Processing SOP, Compliance Checklist, API Documentation (Swagger).

2.7 Assumptions and Dependencies

- Biometric devices support ZKTeco SDK for attendance data pull
- Digio API provisioned for e-signature on offer letters and appointment letters
- Razorpay Payouts or bank NEFT/IMPS integration for salary disbursement
- Statutory rates (PF: 12%, ESI: 3.25% employer / 0.75% employee) configurable in admin
- Income tax slabs (old and new regime) updated by HR admin at start of each financial year

3. External Interface Requirements

3.1 User Interfaces

Interface	Purpose	Technology
Employee Self-Service	Payslip, leave, tax declaration, training	React.js (employee)
HR Console	Payroll, attendance, compliance, analytics	React.js (HR)
Manager Console	Team attendance, leave approval, appraisal	React.js (manager)
Onboarding Portal	Pre-joining document upload, e-sign	React.js (public/semi-public)
Analytics Dashboard	Attrition, headcount, payroll cost	React.js (HR/admin)

3.2 Hardware Interfaces

ZKTeco biometric devices for attendance punch (fingerprint/face); SDK-based data sync. Mobile app (React Native) for geo-fenced attendance punch for field employees.

3.3 Software Interfaces

- ZKTeco SDK — biometric device attendance data pull via SDK; device sync every 15 minutes
- Digio — e-signature API for offer letter, appointment letter, and FnF settlement document
- Razorpay Payouts — salary disbursement API for employee bank accounts; batch payout
- Twilio SMS — OTP, payslip availability notification, and leave approval alerts
- Income Tax Portal (TRACES) — TDS return data export (Form 24Q) in ITD-compliant format

3.4 Communications Interfaces

REST over HTTPS; JWT auth; ZKTeco SDK (proprietary TCP/IP); Razorpay Payout webhook; Digio webhook for e-sign completion; JSON data exchange; PDF for payslips and compliance challans.

4. System Features — Functional Requirements

4.1 FR1: Pre-Joining Onboarding

The system shall manage the complete pre-joining process for new hires from offer letter to Day 1.

- Offer letter generation: HR generates offer letter from template with CTC breakup; sent via email
- E-signature: candidate signs offer letter digitally via Digio; completion webhook triggers next step
- Document checklist: system sends document upload link; candidate uploads ID proof, education, experience
- Document verification: HR marks each document as verified/rejected with remarks
- BGV initiation: HR triggers background verification with vendor (third-party) from portal
- Pre-joining form: candidate fills personal data, bank details, PF UAN, and emergency contacts
- Asset request: HR pre-requests laptop, access card, and email ID before Day 1
- Onboarding kit: automated Day 1 welcome email with portal credentials and handbook PDF
- Probation configuration: probation period set (typically 90 days); probation end alert to HR
- Onboarding tracker: HR dashboard shows onboarding status per new hire with pending checklist items

4.2 FR2: Employee Profile and Organisation

The system shall maintain a comprehensive employee master with organisational hierarchy.

- Employee master: employee ID, name, designation, department, grade, location, and employment type
- Org hierarchy: manager-reportee relationship; org chart auto-generated from hierarchy data
- Employment type: PERMANENT / PROBATION / CONTRACT / INTERN; different policies per type
- Contact details: official email, personal email, mobile, emergency contact, and address
- Bank details: account number, IFSC, bank name — encrypted; used for payroll disbursement
- Statutory IDs: PF UAN, ESI IP number, PAN, Aadhaar (masked display) — stored encrypted
- Document vault: employee stores and views own documents (payslips, letters, certificates)
- Confirmation: probation-to-permanent confirmation action by HR with confirmation letter generation
- Transfer: department or location transfer with effective date; org hierarchy updated
- Employee search: HR searches by name, ID, department, designation, and employment status

4.3 FR3: Attendance Management

The system shall track daily attendance from biometric devices and mobile app with shift scheduling.

- Biometric sync: ZKTeco SDK syncs punch data every 15 minutes; in-punch and out-punch recorded
- Mobile punch: geo-fenced mobile app punch for field employees; GPS coordinates stored
- Shift configuration: HR configures shifts (morning/general/evening/night) with start and end times
- Shift assignment: employee assigned to shift; rotational shift scheduling supported
- Late arrival: late arrival threshold configurable (e.g., 15 minutes); LOP policy applied after N lates
- Half-day policy: first half and second half attendance; one absence = half-day LOP
- Attendance regularisation: employee applies to regularise missed punch; manager approves
- Monthly attendance summary: present, absent, late, and half-day count per employee per month
- Attendance report: downloadable attendance register for payroll processing input
- Holiday calendar: national, state, and company holidays configured; attendance not required

4.4 FR4: Leave Management

The system shall manage leave policies, balances, applications, and approvals.

- Leave types: Casual Leave (CL), Sick Leave (SL), Earned Leave (EL), Maternity, Paternity, Comp-off
- Leave policy: leave quota by employment type and grade; accrual rules (monthly/annually)
- Leave application: employee applies with type, dates, and reason; alternate arrangement noted
- Manager approval: manager approves/rejects/delegates leave; employee notified via SMS and portal
- Leave balance: real-time leave balance displayed per type; year-to-date consumed shown
- Leave encashment: EL encashment at year-end; computed at basic salary per day; included in payroll
- LOP calculation: leaves without balance deducted as LOP in payroll that month
- Comp-off: overtime hours converted to comp-off; employee applies; valid for 90 days
- Leave calendar: team leave calendar visible to manager; prevents too many simultaneous absences
- Leave report: leave utilisation by department, type, and month for HR analytics

4.5 FR5: Payroll Processing

The system shall automate monthly payroll computation with CTC breakup, deductions, and statutory compliance.

- Payroll configuration: CTC components (Basic, HRA, Conveyance, Special Allowance, Bonus) configured per grade
- Monthly run: HR initiates payroll run for a month; system computes gross, deductions, and net pay
- LOP deduction: LOP days from attendance deducted from gross; per-day salary = monthly salary / 26
- PF computation: Employee PF: 12% of Basic + DA; Employer PF: 12% (8.33% EPS + 3.67% EPF)
- ESI computation: applicable for employees with gross > 21,000; Employee: 0.75%; Employer: 3.25%
- TDS computation: monthly TDS based on projected annual income, declared investments, and tax regime
- Professional Tax: PT deducted per state slab; configurable by state and salary range
- Net pay: Gross - (Employee PF + ESI + TDS + PT + Loan EMI + LOP); transferred via Razorpay Payouts
- Payroll approval: HR reviews payroll summary; CFO/Admin approves before disbursement
- Payroll lock: approved payroll locked; no changes permitted; revision requires next-month adjustment

4.6 FR6: Payslip and Statutory Compliance

The system shall generate payslips and statutory challans for PF, ESI, TDS, and PT compliance.

- Payslip PDF: monthly payslip with earnings, deductions, and net pay; password-protected (DOB DDMMYYYY)
- Payslip delivery: emailed to official email; available for download in employee self-service portal
- YTD summary: year-to-date earnings and deductions displayed on payslip for income tax reference
- Form 16: annual Form 16 (Part A and Part B) generated per employee for income tax filing
- PF challan: monthly ECR (Electronic Challan cum Return) file generated in EPFO-prescribed format
- ESI challan: monthly ESI challan in ESIC-prescribed format for portal upload
- TDS return: quarterly Form 24Q data export in ITD format for TDS return filing
- PT challan: state-specific PT challan generated per applicable states
- Arrear payroll: arrear computation when salary revision is applied with back-effective date
- Investment declaration: employees submit Form 12BB; system uses to compute monthly TDS

4.7 FR7: Performance Management

The system shall manage the full performance appraisal cycle from goal setting to final rating.

- Goal setting: employee sets KRAs/KPIs at start of appraisal year; manager reviews and approves goals
- Mid-year review: structured mid-year check-in form; employee self-assessment; manager feedback
- Annual appraisal: employee self-assessment; manager rating; skip-level review; HR normalisation
- Rating scale: 1–5 scale (5: Outstanding, 4: Exceeds Expectations, 3: Meets, 2: Partial, 1: Below)
- Appraisal workflow: Employee Manager Skip Level HR Normalisation Final Rating
- Bell curve: HR applies bell curve normalisation across department to maintain rating distribution
- Increment computation: configured increment % per rating band applied to basic salary
- Appraisal letter: increment and revised CTC letter generated and e-signed via Digio
- PIP: Performance Improvement Plan initiated for rating 2; manager defines PIP goals and timeline
- Appraisal analytics: rating distribution by department; average rating trend year-on-year

4.8 FR8: Training and Development

The system shall manage training programmes, nominations, attendance, and completion certification.

- Training calendar: HR publishes training sessions with title, date, mode (online/offline), and seats
- Nomination: employee nominates self or manager nominates team member for training
- Approval: manager approves nomination; employee notified with joining instructions
- Attendance marking: trainer marks attendance; absent employees flagged to HR
- Completion certificate: on 100% attendance, system generates and emails certificate to employee
- Training MIS: training hours per employee; training cost; completion rate per programme
- Mandatory training: compliance training (POSH, fire safety) assigned to all; completion tracked
- External training: employee logs externally attended training for record; manager approves
- Training calendar view: employees browse upcoming training and self-nominate with manager approval
- Training effectiveness: post-training survey with rating; results linked to training record

4.9 FR9: Employee Self-Service

The system shall provide employees with a self-service portal for all HR transactions.

- Payslip access: employee downloads payslips for any month in the current and previous financial year
- Leave application: employee applies leave from self-service; tracks approval status in real time
- Tax declaration: employee submits Form 12BB investment declaration; updates throughout the year
- Profile update: employee updates personal mobile, emergency contact, and address; bank update needs HR approval
- IT declaration: new and old tax regime selection; system recalculates TDS on selection change
- Help desk: employee raises HR queries (letter request, policy clarification); resolved within 2 days
- Attendance regularisation: employee regularises missed punch with reason and alternate proof
- Loan application: employee applies for salary advance or interest-free loan; HR/Finance approves
- Document requests: employee requests experience letter, salary certificate, Form 16 from portal
- Policy hub: HR publishes company policies (leave, travel, POSH); employees acknowledge reading

4.10 FR10: Exit Management

The system shall manage the complete employee exit process from resignation to FnF settlement.

- Resignation submission: employee submits resignation from self-service with last working day (LWD)
- Notice period computation: system calculates notice period from designation grade; LWD confirmed
- Manager approval: manager acknowledges resignation; counter-offer flag if applicable

- Exit interview: structured exit interview form sent to employee; responses confidential to HR
- Clearance workflow: parallel clearance from IT (laptop return), Admin (access card), Finance (loans)
- Clearance dashboard: HR tracks pending clearances from each department with due dates
- FnF computation: earned but unused EL encashment + pending salary - recoveries (loan, notice period shortfall)
- FnF payslip: dedicated FnF payslip with all components and statutory deductions for last month
- Experience letter: auto-generated on clearance completion; e-signed via Digio
- PF and ESI exit: PF withdrawal / transfer form pre-filled; employee downloads and submits to EPFO

4.11 FR11: HR Analytics and Reporting

The system shall provide data-driven HR analytics and standard statutory reports.

- Headcount dashboard: live headcount by department, location, gender, and employment type
- Attrition analytics: monthly attrition rate; voluntary vs involuntary; department-wise trend
- Payroll cost: total payroll cost (CTC) by department and grade; month-on-month trend
- Leave utilisation: leave consumed vs balance by type; department-wise leave pattern
- Attendance analytics: average attendance rate; late arrival trend; LOP impact on payroll
- Appraisal distribution: rating distribution by department and grade; bell curve compliance
- Training hours: training hours per employee; mandatory training compliance percentage
- New joiners vs exits: monthly joining and exit count; net headcount change trend
- Statutory compliance calendar: PF/ESI/TDS due dates with filing status (FILED/PENDING/OVERDUE)
- Custom report: HR generates ad-hoc reports with any combination of filters; export Excel/PDF

4.12 FR12: Admin Configuration and Access Control

The system shall allow HR admin to configure all HR policies, statutory rates, and system access.

- Company configuration: company name, registered address, PF/ESI registration numbers, GSTIN
- Grade and band: define job grades with CTC ranges, leave quotas, and notice period mapping
- Statutory rates: configure PF%, ESI%, PT slabs per state, and income tax slabs per regime
- Leave policy engine: configure leave quotas, accrual rules, carry-forward, and encashment per grade
- Payroll calendar: configure payroll cut-off date, processing date, and disbursement date per month
- RBAC management: create HR manager, department manager accounts; assign role and scope
- Audit log: all payroll actions, configuration changes, and data access logged immutably
- Data export: full employee master, attendance, payroll, and statutory data export for audits
- Notification templates: customise all system-generated emails and SMS messages
- Financial year rollover: year-end leave carry-forward, EL encashment, and TDS reset for new FY

5. Other Non-Functional Requirements

5.1 Performance Requirements

Metric	Target	Verification
Payroll Run (5,000 employees)	< 2 hours	Benchmark
Payslip PDF Generation	< 5 seconds per employee	Performance test
Attendance Sync (biometric)	< 15 minutes lag	Integration test
Leave Application to Approval	< 30 seconds after manager action	Load test
Concurrent Self-Service Users	2,000 simultaneous	Load test
API Response (P95)	< 400ms	Performance test
Form 16 Generation	< 10 minutes for 1,000 employees	Benchmark
PF ECR File Generation	< 5 minutes	Benchmark
System Uptime	99.5%	Uptime monitoring
Report Generation	< 30 seconds	Performance test

5.2 Safety Requirements

All data mutations require authenticated JWT session. Destructive operations (delete, bulk update) require secondary confirmation. Audit logs capture all privileged actions.

5.3 Security Requirements

- TLS 1.2+ enforced on all endpoints; HTTP redirected to HTTPS
- JWT HS512 with 1-hour access token; refresh token rotation on use
- RBAC: role validated server-side on every request; never client-controlled
- Passwords hashed with BCrypt (cost factor 12); never stored in plain text
- SQL injection prevention via JPA parameterised queries exclusively
- XSS prevention: Content-Security-Policy header; React auto-escaping
- CORS: restricted to registered frontend origins only
- VAPT conducted annually; critical vulnerabilities patched within 72 hours

5.4 Software Quality Attributes

JUnit 5 unit and integration tests with 80% code coverage. CI/CD via GitHub Actions. SonarQube for static analysis. Flyway for database migrations.

5.5 Database Requirements

Table	Purpose	Key Fields
employees	Employee master	employee_id (PK UUID), emp_code (VARCHAR 20 UNIQUE), name (VARCHAR 100), de
salary_structures	CTC component config	per_gradestructure_id (PK UUID), grade (VARCHAR 20), effective_from (DATE), basic_pct
payroll_runs	Monthly payroll run	headerrun_id (PK UUID), run_month (DATE), status (ENUM: DRAFT/PENDING_APPROVAL/A

payroll_records	Per-employee payroll detail	record_id (PK UUID), run_id (FK), employee_id (FK), gross (DECIMAL 10,2), basic (DE
attendance_records	Daily attendance	attendance_id (PK UUID), employee_id (FK), work_date (DATE), in_punch (TIME nullable
leave_applications	Leave requests	leave_id (PK UUID), employee_id (FK), leave_type (VARCHAR 30), from_date (DATE), to
leave_balances	Employee leave balance per type	balance_id (PK UUID), employee_id (FK), leave_type (VARCHAR 30), financial_
appraisals	Performance appraisal records	appraisal_id (PK UUID), employee_id (FK), appraisal_year (VARCHAR 9), employee
exit_records	Employee exit management	exit_id (PK UUID), employee_id (FK UNIQUE), resignation_date (DATE), last_working
statutory_config	Statutory rate configuration	config_id (PK UUID), config_key (VARCHAR 100 UNIQUE), config_value (TEXT), effect
training_records	Training attendance and completion	training_id (PK UUID), employee_id (FK), programme_name (VARCHAR 200)
hr_audit_logs	HR system audit trail	log_id (PK UUID), actor_id (UUID), action (VARCHAR 200), entity_type (VARCHAR 100)

5.6 System Quality Attributes

3.6.1 Reliability

99.5% uptime; payroll run retry on failure; biometric sync queue with retry; no data loss guarantee.

3.6.2 Availability

24×7 self-service; payroll processing window: 1st–5th of month; peak load during 10 AM–12 PM IST.

3.6.3 Security

TLS 1.2+; JWT HS512; RBAC; AES-256 for PAN/Aadhaar; payslip password-protected; annual VAPT.

3.6.4 Maintainability

JUnit 5 80% coverage; Flyway; Swagger; SonarQube; statutory rate config externalised (not hardcoded).

3.6.5 Portability

Docker + Kubernetes; cloud-agnostic; database-agnostic JPA; React.js cross-browser (Chrome/Firefox/Safari).

5.7 Design Constraints

- Payroll is a two-step process: COMPUTE → APPROVE → DISBURSE; no single-step disbursement
- Payslips are password-protected PDF; password is employee DOB in DDMMYYYY format
- PAN and Aadhaar stored encrypted (AES-256-GCM); masked on all UI displays
- Biometric fingerprint templates stored on device only; never transmitted to application server
- Payroll data visible only to HR Manager and Finance roles; RBAC enforced at API level
- Spring Boot 3.x, Java 17+; React.js functional components; Flyway for DB migrations
- HTTPS-only; JWT auth; RBAC; all salary data encrypted in transit; audit log immutable
- FnF computation must account for all unpaid salary, LOP, EL encashment, and loan recovery

5.8 Other Requirements

EPF Act 1952: PF contributions and ECR filing compliance. ESI Act 1948: ESI contributions and challan. IT Act 1961: TDS deduction and Form 24Q return. Companies Act 2013: statutory registers maintained. POSH Act: mandatory POSH training tracked. Data retention: payroll records 7 years; attendance 3 years.

Appendix A: Permission Matrix

Feature	Super Admin	HR Manager	Dept Manager	Employee
Run Payroll				

lapsed: DECIMAL(5,1) NOT NULL DEFAULT 0
UNIQUE(employee_id, leave_type, financial_year)

B.3 exit_records Table

exit_id: UUID PRIMARY KEY DEFAULT gen_random_uuid()
employee_id: UUID NOT NULL UNIQUE REFERENCES employees
resignation_date: DATE NOT NULL
last_working_day: DATE NOT NULL
exit_type: ENUM NOT NULL
clearance_status: JSONB NOT NULL DEFAULT '{}'
fnf_amount: DECIMAL(10,2) nullable
fnf_status: ENUM NOT NULL DEFAULT 'PENDING'
exit_interview_submitted: BOOLEAN NOT NULL DEFAULT FALSE

Appendix C: Validation Rules

Field	Rule	Error Message
PAN Number	Format: AAAAA9999A (5 alpha + 4 numeric + 1 alpha)	Invalid PAN format
Payroll Run	Only one active payroll run per month allowed	Payroll already initiated for this month
Leave Application	From date must be future or today; balance must be sufficient	Insufficient leave balance
LWD (Exit)	Last working day = resignation date + notice period days	Notice period not completed
PF Wage Ceiling	PF computed on actual basic or ₹ 15,000 cap, whichever is lower	PF wage ceiling applied
ESI Eligibility	Applicable only for employees with gross salary ≤ ₹ 21,000	ESI not applicable for this salary range
FnF Computation	All clearances must be CLEARED before FnF payment	Pending clearances — complete before FnF
Appraisal Rating	1.0 to 5.0 in 0.5 increments	Rating must be between 1.0 and 5.0
Attendance Regularisation	Must be applied within 7 days of the punch date	Regularisation window expired
TDS Computation	Based on employee's chosen tax regime (old/new)	Tax regime not selected — defaulting to new regime

Appendix D: Security Configuration and Exception Handling

Parameter	Value
Algorithm	HS512
Access Token Expiry	3600s
Payroll Data Access	Restricted to HR_MANAGER and FINANCE roles; enforced at API gateway
PAN/Aadhaar Encryption	AES-256-GCM; key in AWS KMS; never in application config
Payslip Password	Employee DOB in DDMMYYYY format; set at PDF generation; not stored
Audit Log	Immutable; INSERT-only table; DB trigger prevents UPDATE/DELETE
Rate Limiting	Payroll run: admin-only; leave applications: 5 per employee per day

D.2 Global Exception Handling

- GlobalExceptionHandler — RFC 7807 problem+json
- PayrollAlreadyRunException — duplicate payroll run for same month (HTTP 409)
- InsufficientLeaveBalanceException — leave application without balance (HTTP 400)
- ClearancePendingException — FnF attempt with pending clearances (HTTP 400)
- PayrollLockedException — edit attempt on approved payroll (HTTP 403)
- InvalidTaxRegimeException — invalid regime selection (HTTP 400)
- BiometricSyncFailureException — ZKTeco SDK sync failure (handled internally with retry)
- StatutoryRateNotFoundException — no config found for applicable PT state/ESI (HTTP 500 with alert)

Appendix E: Integration Specifications

E.1 ZKTeco Biometric SDK

Attendance data pulled from ZKTeco biometric devices via SDK over local network every 15 minutes. In-punch and out-punch logs synced to attendance_records table. Device time synced to NTP server to prevent clock drift. Missed sync triggers retry with 5-minute backoff.

E.2 Digio e-Sign

Offer letters, appointment letters, increment letters, and FnF settlements e-signed via Digio API. Document uploaded to Digio; signing link sent to employee email/mobile. Webhook triggers on signing completion; signed PDF stored in S3 and linked to employee record.

E.3 Razorpay Payouts

Salary disbursement via Razorpay Payouts API after payroll approval. Batch payout created with employee account number, IFSC, and net pay. Payout status webhook updates disbursement_status in payroll_records. Failed payouts flagged to HR for manual NEFT processing.

Appendix F: Business Rules

F.1 Payroll Rules

- Payroll run for a month is allowed only between 25th of current month and 10th of next month
- LOP deduction = (Monthly Gross / 26) × LOP days; applied before any other deduction
- PF computed on Basic + DA capped at ₹ 15,000; excess basic not included in PF basis
- Payroll approved by HR Manager + Finance approval; disbursement only after dual approval

F.2 Leave and Exit Rules

- Earned Leave encashment at year-end: maximum carry-forward configured per grade; excess encashed
- Notice period buyout: employee can pay salary-in-lieu to exit early; HR Finance approval required
- FnF computation completed within 45 days of LWD per Payment of Wages Act
- Absconding employee: exit type ABSCONDING; no FnF; PF transferred to EPFO unclaimed account

Appendix G: Future Roadmap

Phase 2 (Q3 2026)

- AI-based attendance anomaly detection: flag suspicious geo-fence punches
- Mobile ESS app: dedicated React Native app for employee self-service

Phase 3 (Q1 2027)

- AI performance insights: natural language performance review summaries from structured ratings
- WhatsApp self-service: payslip and leave balance via WhatsApp bot

Phase 4 (Q3 2027)

- Workforce planning: headcount forecasting and budget planning module
- Multi-company support: conglomerate HR with entity-wise payroll and shared employee pool

Appendix H: API Summary

H.1 Employee and Attendance APIs

Method	Endpoint	Description	Role
POST	/api/v1/employees	Create employee profile	HR Manager
GET	/api/v1/employees/{id}	Get employee details	HR/Manager
POST	/api/v1/attendance/regularise	Submit regularisation	Employee
GET	/api/v1/attendance/{month}	Monthly attendance summary	HR/Manager
GET	/api/v1/leave/balance	Employee leave balance	Employee

H.2 Payroll and Leave APIs

Method	Endpoint	Description	Role
POST	/api/v1/payroll/run	Initiate payroll run	HR Manager
POST	/api/v1/payroll/{id}/approve	Approve payroll	HR Manager
GET	/api/v1/payroll/payslip/{month}	Download payslip	Employee
POST	/api/v1/leave/apply	Apply for leave	Employee
PUT	/api/v1/leave/{id}/approve	Approve leave	Manager

H.3 Compliance and Exit APIs

Method	Endpoint	Description	Role
GET	/api/v1/compliance/pf-ecr/{month}	Generate PF ECR file	HR Manager
GET	/api/v1/compliance/form16/{year}	Generate Form 16	HR Manager
POST	/api/v1/exit/resign	Submit resignation	Employee
GET	/api/v1/exit/fnf/{id}	Compute FnF settlement	HR Manager
GET	/api/v1/analytics/headcount	Headcount dashboard	HR Manager

Appendix I: React Component Structure

I.1 App.jsx

Redux store (employee session, leave balance, payroll state, notification queue), React Router v6 with role guards (employee/manager/HR), JWT interceptor, payroll run state machine, financial year context.

I.2 NavBar.jsx

Logo, employee name and ID, leave balance quick-view widget, payslip alert badge, notification bell (leave approvals, training reminders), help desk shortcut, profile dropdown.

I.3 Home.jsx

Employee home: quick-action tiles (Apply Leave, Download Payslip, Tax Declaration, Raise Query), attendance punch widget, pending approvals for managers, payroll run status for HR, compliance calendar.

I.4 HRManagementPlatform.jsx

Onboarding Flow: offer letter generation, e-sign tracking, document checklist, BGV status, asset request. Employee Directory: org chart, profile viewer, search. Attendance: daily punch log, regularisation form, leave calendar. Leave Module: application form, approval queue, balance display. Payroll: run wizard (compute review approve disburse), payslip viewer, statutory file downloads. Appraisal: goal sheet, self-assessment form, rating dashboard, bell curve view. Training: calendar, nomination, attendance marking, certificate. Exit: resignation form, clearance tracker, FnF summary. Analytics: headcount, attrition, payroll cost, leave charts.

I.5 Footer.jsx

Company name, HR helpdesk contact, privacy policy, statutory compliance note, grievance officer contact (POSH), copyright.

Version Control History

Version	Date	Author	Description
1.0	2026-06-16	System Architect	Initial draft — complete SRS